

FROM THE VIRIAL TO THE CUBIC EQUATION OF STATE *

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ABSTRACT

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Second virial coefficients are used extensively in low-pressure vapor–liquid equilibrium calculations to predict the vapor-phase fugacity coefficient. Tsonopoulos's 1974 B (second virial coefficient) correlation is reviewed, with particular emphasis on the dependence of the polar parameter a on the reduced dipole moment. New results for the B of water are analyzed and the 1974 recommendation is discarded in favor of $a = -0.0109$. New data also make it possible to correlate the characteristic constant k_{ij} for water/ n -alkane binaries with the critical volume of the hydrocarbon. For calculations at reduced densities significantly higher than 0.25, B is combined with the Redlich–Kwong equation to form the “Virial-RK” equation of state. This equation predicts reasonably good estimates for the third virial coefficient of non-polar gases, such as methane, but the predictions for water are very poor. However, the fugacity coefficients predicted by virial-RK for the water/methane binary are reliable up to very high pressures, except in the vicinity of the critical point.

INTRODUCTION

VLE (vapor–liquid equilibrium) calculations are usually carried out, at least in industry, with various forms of the equation of state. One reason for the popularity of equations of state, in comparison with hybrid models, is the great simplification which they introduce into the calculations.

The starting point of all VLE calculations is the equilibrium condition

$$f_i^V = f_i^L \quad (1)$$

That is, the fugacity of component i in the vapor should be equal to its

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fugacity in the liquid. If the fugacity is expressed in terms of the fugacity coefficient, eqn. (1) becomes

$$\phi_i^V y_i P = \phi_i^L x_i P \quad (2)$$

and the following expression for the K value results:

$$K_i = \frac{y_i}{x_i} = \frac{\phi_i^L}{\phi_i^V} \quad (3)$$

Equation (3) provides the simplest and most direct method for VLE calculations—it requires no standard states or special procedures as the critical point is approached—provided the equation of state applies to both phases. Some equations do well, if we exclude strongly polar compounds and electrolytes.

For mixtures containing strongly polar compounds or electrolytes, VLE calculations are commonly carried out with hybrid models; that is, models which use activity coefficients for the liquid and fugacity coefficients for the vapor, which are calculated with an equation of state. In that case, eqn. (2) is replaced by

$$\phi_i^V y_i P = \gamma_i x_i f_i^0 \quad (4)$$

where γ_i is the activity coefficient and f_i^0 is the reference fugacity. It follows that the expression for the K value now becomes

$$K_i = \frac{\gamma_i f_i^0}{\phi_i^V P} \quad (5)$$

In this paper we are only concerned with the calculation of ϕ_i^V . At densities up to one quarter of the critical density (roughly speaking, up to pressures of 15 bar at subcritical conditions), ϕ_i^V can be reliably calculated with the virial equation of state truncated after the second term.

$$Z = \frac{Pv}{RT} = 1 + \frac{B}{v} \quad (6)$$

A great advantage of the virial equation is that it can be extended to mixtures without arbitrary assumptions. For a mixture of n components, the mixture second virial coefficient, B_m , is given exactly by

$$B_m = \sum_{i=1}^n \sum_{j=1}^n y_i y_j B_{ij} \quad (7)$$

B_{ij} ($i \neq j$) is the second virial cross-coefficient; it is a function only of temperature, like the pure-component coefficients (B_{ii} and B_{jj}). It plays a

pre-eminent role in the calculation of ϕ_i^V :

$$\ln \phi_i^V = \frac{2}{v_m} \sum_{j=1}^n y_j B_{ij} - \ln Z_m \quad (8)$$

In the limit as component 1 becomes infinitely dilute in 2, if we consider only a binary, eqn. (8) becomes

$$\ln \phi_1^{\infty,V} = \frac{2B_{12}}{v_2} - \ln Z_2 \quad (9)$$

Thus, a reliable estimate of the cross-coefficient B_{12} is essential in the calculation of fugacity coefficients. Both pure-component B s and cross-coefficients can be predicted with several correlations available in the literature. Here we will use only the correlation of Tsonopoulos (1974) and will update it for water and water/*n*-alkane binaries.

Although eqn. (8) is a very useful relationship, it is limited to relatively low densities or pressures. For high-pressure VLE calculations, we combine B with the Redlich-Kwong (1949) equation of state, and show how this equation can be used to predict fugacity coefficients for mixtures.

THE TSONOPOULOS B CORRELATION

In reduced form, the Tsonopoulos (1974) correlation is written as

$$\frac{BP_c}{RT_c} = f^{(0)}(T_r) + \omega f^{(1)}(T_r) + f^{(2)}(T_r) \quad (10)$$

where

$$f^{(0)}(T_r) = 0.1445 - 0.330/T_r - 0.1385/T_r^2 - 0.0121/T_r^3 - 0.000607/T_r^8 \quad (11)$$

$$f^{(1)}(T_r) = 0.0637 + 0.331/T_r^2 - 0.423/T_r^3 - 0.008/T_r^8 \quad (12)$$

$$f^{(2)}(T_r) = a/T_r^6 - b/T_r^8 \quad (13)$$

$f^{(2)}(T_r)$ is the polar term; it is important at $T_r < 1$, but reduces rapidly to zero with increasing T_r . For non-hydrogen-bonding polar compounds such as ketones, it is only necessary to use a ; that is, $b = 0$. Since the B of polar compounds is more negative than that of non-polar compounds (for the same T_r and ω), it follows that $f^{(2)} < 0$ or $a < 0$. However, the $f^{(2)}$ of dimerizing compounds such as alkanols has a more complex temperature dependence: it is steeper at very low T_r (< 0.6), but is flatter at $T_r > 0.8$. As a result, both parameters (which assume positive values) in eqn. (13) must be used. For a recent re-examination of the B of alkanols, see Tsonopoulos et al. (1989).

Dependence of a on reduced dipole moment

Tsonopoulos (1974) demonstrated that the polar parameter a can be correlated with μ_r , the reduced dipole moment:

$$\mu_r = 10^5 \mu^2 P_c / T_c^2 \quad (14)$$

(In eqn. (14), μ is given in debyes (1 debye = 3.33567×10^{-30} C · m), P_c in atmospheres (1 atm = 0.101325 MPa), and T_c in kelvins.) However, it was also shown that, although a must go to zero as μ_r goes to zero, a cannot be given for all polar compounds by a unique function of μ_r .

Figure 1 summarizes what has been found (Tsonopoulos, 1974, 1975, 1978) for the dependence of a on μ_r . The top curve (1) in Fig. 1 is recommended for ketones, aldehydes, alkyl nitriles, ethers, and carboxylic acid esters.

$$a = -2.14 \times 10^{-4}(\mu_r) - 4.308 \times 10^{-21}(\mu_r)^8 \quad (15)$$

However, for alkyl halides, mercaptans, sulfides, and disulfides, the bottom curve (2) in Fig. 1 should be used.

$$a = -2.188 \times 10^{-11}(\mu_r)^4 - 7.831 \times 10^{-21}(\mu_r)^8 \quad (16)$$

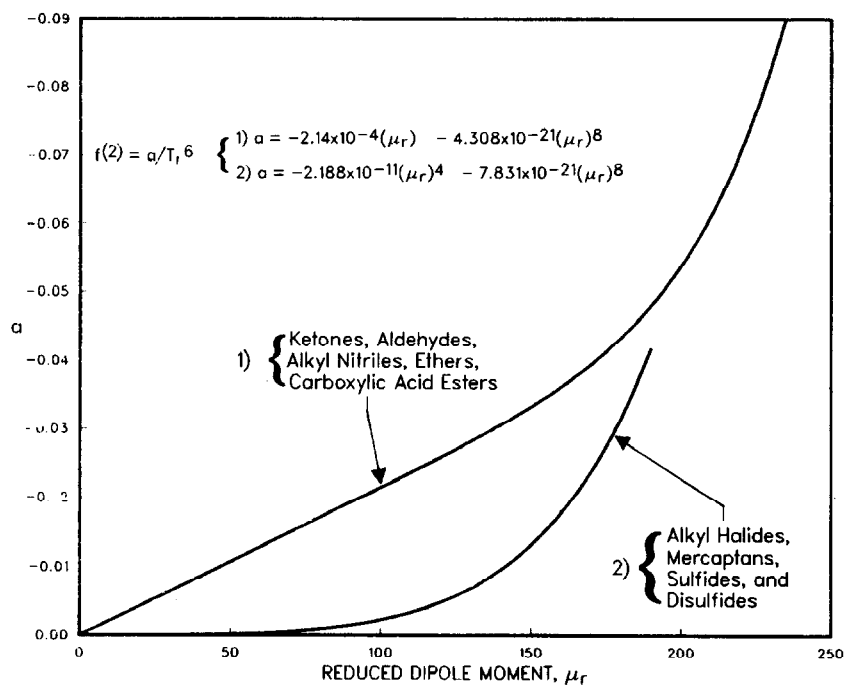


Fig. 1. Dependence of a on reduced dipole moment.

Equations (15) and (16) present a satisfactory correlation for most non-dimerizing polar organic compounds. These two equations approach closely at high μ_r , suggesting that a unique relationship between a and μ_r is possible only for strongly polar, but non-dimerizing, organic compounds.

Second virial coefficient of water

Figure 2 presents all the B data for water reported by Cholinski et al. (1986), along with the recent measurements of Eubank et al. (1988), who reported two sets of data. Also shown in Fig. 2 are the calculations with the 1974 correlation; first, with $f^{(2)} = 0$ (i.e. without a polar term), then with the 1974 recommendation

$$f^{(2)} = 0.0279/T_r^6 - 0.0229/T_r^8 \quad (17)$$

and finally with the revised recommendation

$$f^{(2)} = -0.0109/T_r^6 \quad (18)$$

The predictions with $f^{(2)} = 0$ are as expected: the calculated B is too positive. However, the 1974 recommendation for $f^{(2)}$ (eqn. (17)) makes B

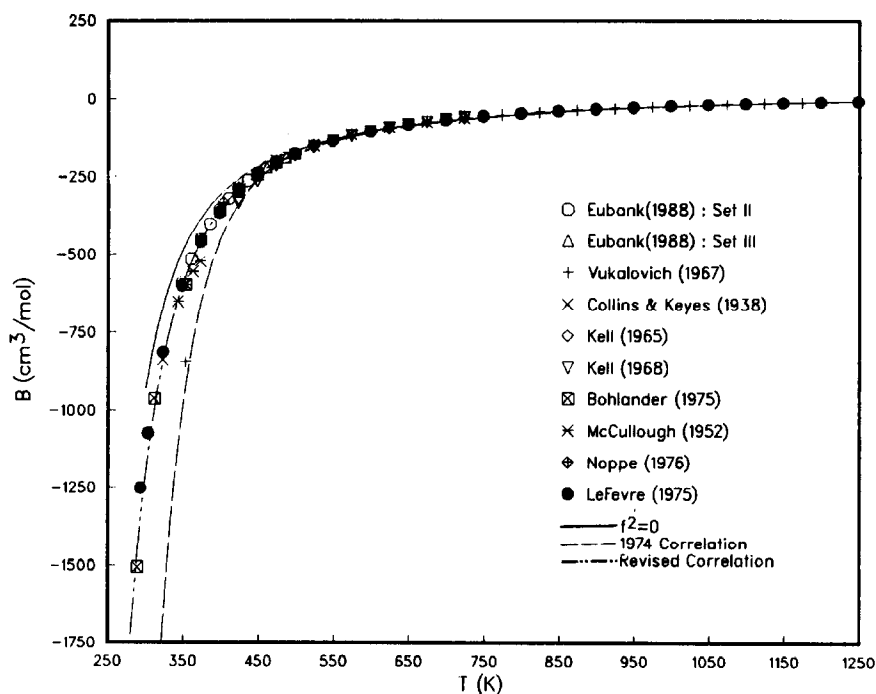


Fig. 2. Second virial coefficient of water. All references are in Cholinski et al. (1986), except for Eubank et al. (1988).

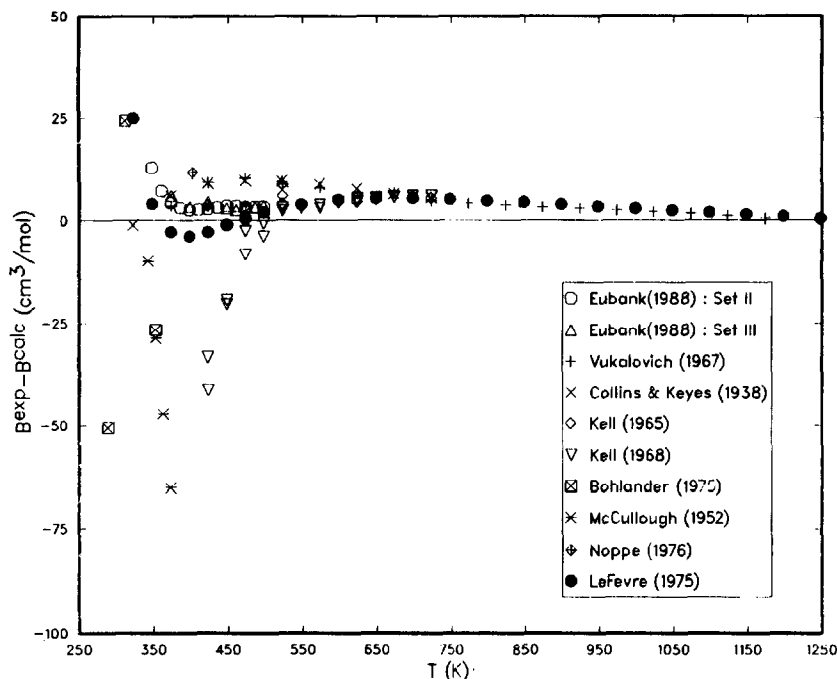


Fig. 3. Second virial coefficient of water. Deviation from revised correlation: $f^{(2)} = -0.0109/T_r^6$. All references are in Cholinski et al. (1986), except for Eubank et al. (1988).

far too negative at low temperatures. This is because the 1974 recommendation was based, at low temperatures, on the data of Kell et al. (1968). These data have been shown, most recently and conclusively by Eubank et al. (1988), to be too negative owing to significant adsorption effects that had not been accounted for.

The problem with the data of Kell et al. (1968) is clearly shown in Fig. 3, where the difference between experimental and calculated (revised recommendation, eqn. (18)) B values is plotted vs. temperature. The data of Kell et al. (1968), along with the 353 K value from Vukalovich et al. (1967) (see Fig. 2), suggest a much steeper dependence on temperature, which is not supported by the recent results of Eubank et al. (1988), as well as of others.

Equation (18), the revised recommendation for water, fits all 118 data points with an average deviation of $11.0 \text{ cm}^3 \text{ mol}^{-1}$. As shown in Fig. 3, the deviations exceed $25 \text{ cm}^3 \text{ mol}^{-1}$ only below 450 K.

CROSS-COEFFICIENTS FOR MIXTURES

The second virial cross-coefficient, B_{ij} , has the same temperature dependence that B_{ii} and B_{jj} have, but the parameters to be used with eqns.

(10)–(13) are P_{cij} , T_{cij} , ω_{ij} , a_{ij} , and b_{ij} . The following mixing rules make it possible to relate these characteristic constants to pure-component parameters.

$$T_{cij} = (T_{ci}T_{cj})^{1/2}(1 - k_{ij}) \quad (19)$$

$$P_{cij} = \frac{4T_{cij}(P_{ci}v_{ci}/T_{ci} + P_{cj}v_{cj}/T_{cj})}{(v_{ci}^{1/3} + v_{cj}^{1/3})^3} \quad (20)$$

$$\omega_{ij} = 0.5(\omega_i + \omega_j) \quad (21)$$

where k_{ij} is a characteristic constant for each binary.

Equations (19)–(21) suffice for non-polar/non-polar binaries. For polar/non-polar binaries, B_{ij} is assumed to have no polar term:

$$a_{ij} = 0 \quad (22)$$

$$b_{ij} = 0 \quad (23)$$

For polar/polar binaries, the polar contribution to B_{ij} is calculated by assuming that

$$a_{ij} = 0.5(a_i + a_j) \quad (24)$$

$$b_{ij} = 0.5(b_i + b_j) \quad (25)$$

The most sensitive mixing rule is eqn. (19). T_{cij} can be assumed to be the geometric mean of T_{ci} and T_{cj} ($k_{ij} = 0$) only when i and j are very similar in size and chemical nature. Otherwise, in the absence of any specific chemical interaction between i and j , k_{ij} should be positive and thus T_{cij} would be less than the geometric mean.

CROSS-COEFFICIENTS FOR WATER/n-ALKANE BINARIES

Tsonopoulos (1974, 1975, 1978) has analyzed cross-coefficient data for a wide variety of polar binaries. In every case such analysis involves the optimization of the characteristic binary constant k_{ij} . Correlations for the k_{ij} of non-polar and polar systems were presented by Tsonopoulos (1979) and, for alkane/alkane and alkane/alkanol binaries, by Tsonopoulos et al. (1989). Here we consider only the binaries of water with n-alkanes.

Tsonopoulos (1974) recommended for water/methane and water/ethane, respectively, k_{ij} values of 0.34 and 0.37. These are plotted as solid circles in Fig. 4 (k_{ij} vs. the critical volume of the alkane). An average k_{ij} value of 0.40 was suggested for other water/hydrocarbon binaries, although it was later found that k_{ij} can be correlated with the critical volume of the hydrocarbon (Tsonopoulos, 1979).

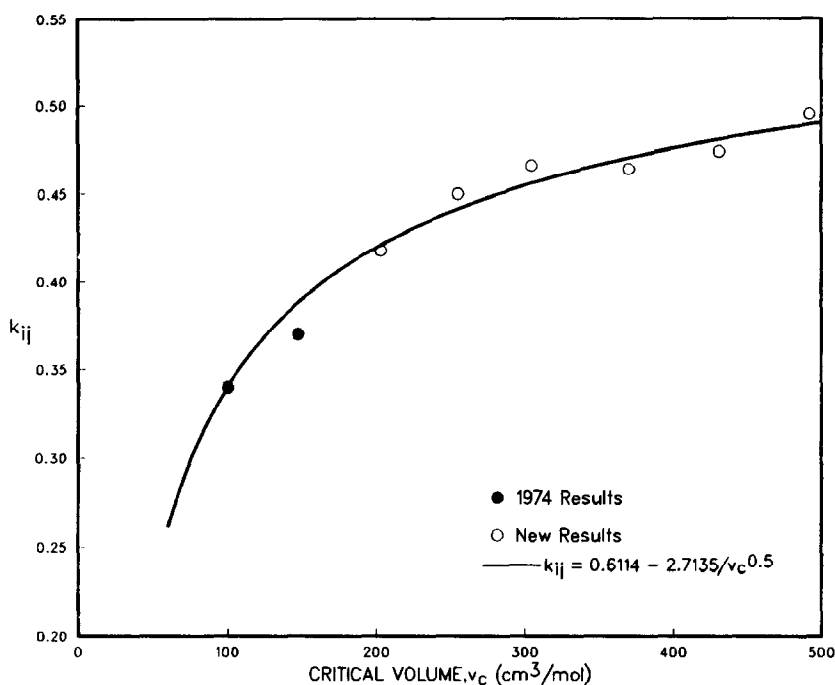


Fig. 4. Optimum k_{ij} values for water/n-alkane binaries.

These early results have been confirmed by analyzing B_{ij} values derived from the excess enthalpy measurements of Wormald and co-workers. The results for C_3 – C_8 n-alkanes are given in Table 1 and plotted in Fig. 4, which

TABLE 1

Optimum k_{ij} values for water/n-alkane binaries

Alkane	k_{ij}	Data sources and comments
Methane	0.34	see Tsonopoulos (1974); confirmed by data of Joffrion and Eubank (1988)
Ethane	0.37	see Tsonopoulos (1974); confirmed by data of Lancaster and Wormald (1985)
Propane	0.418	Skripka (1979); Lancaster and Wormald (1986); Wormald and Lancaster (1986)
Butane	0.450	Lancaster and Wormald (1986); Wormald and Lancaster (1986)
Pentane	0.466	Wormald et al. (1983); Smith et al. (1984)
Hexane	0.463	Skripka (1979); Wormald et al. (1983); Smith et al. (1984)
Heptane	0.473	Richards et al. (1981); Wormald et al. (1983); Smith et al. (1984)
Octane	0.495	Skripka (1979); Wormald et al. (1983); Smith et al. (1984)

also shows that a very simple relationship fits the k_{ij} values for all water/n-alkane binaries:

$$k_{ij} = 0.6114 - 2.7135/v_c^{0.5} \quad (26)$$

Equation (26) should also provide a good approximation for all water/non-polar gas binaries.

VIRIAL EQUATION OF STATE

The virial equation truncated after the B term is limited to densities less than one quarter the critical density or pressures lower than 15 bar (at subcritical conditions). To go to higher densities or pressures, we need C , the third virial coefficient, and even higher virial coefficients.

The density series

$$Z = 1 + \frac{B}{v} + \frac{C}{v^2} + \dots \quad (27)$$

is superior to the pressure series

$$Z = 1 + B'P + C'P^2 + \dots \quad (28)$$

where $B' = B/RT$ and $C' = (C - B^2)/(RT)^2$; for example, see Prausnitz et al. (1986). Equation (27) can be used with confidence up to one half the critical density. Lee et al. (1978, 1979) have presented graphically the reduced pressure and temperature ranges that can be described by eqns. (27) or (28) with B , B and C , or even higher virial coefficients. However, even an infinite series cannot describe accurately the critical region. What is more important is that so little is known about C , especially for polar systems, that in general the virial equation is truncated after the B (or B') term.

When only the second virial coefficient is used, it is important to differentiate between $T < T_c$ and $T > T_c$. At supercritical temperatures, the truncated pressure series is superior to the truncated density series. Indeed, for non-polar gases, $C' \approx 0$ for $T_r > 1.4$ (Chueh and Prausnitz, 1967), and therefore the truncated pressure series is equivalent to eqn. (28).

What is not so well known is that the truncated density series is the superior form at subcritical temperatures. This is clearly illustrated in Fig. 5. As shown, the truncated density series predicts reliably the compressibility factor of saturated steam up to $T_r \approx 0.9$, but then it breaks down, predicting complex values at $T_r > 0.925$. In contrast, the truncated pressure series is reasonably good only up to $T_r \approx 0.8$.

In order to go to $T_r > 0.9$, we must use either C or a closed-form equation of state, such as a cubic equation. However, most cubic equations give poor results for polar gases. This limitation can be drastically diminished if we

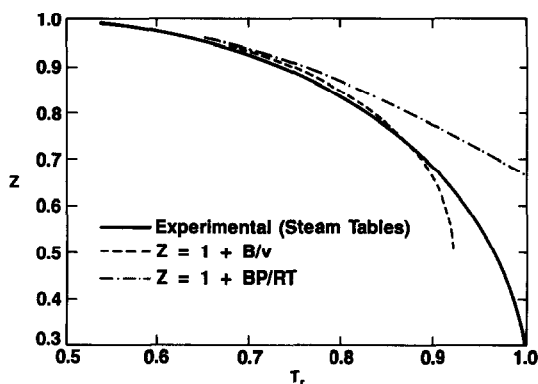


Fig. 5. Compressibility factor of saturated steam.

combine the B correlation presented earlier with a cubic equation. This combination is presented in the next section.

VIRIAL-RK EQUATION OF STATE

The cubic equation of choice is that of Redlich and Kwong (1949), because it has already been shown (Tsonopoulos and Heidman, 1985) that the Redlich-Kwong equation is in better agreement with the B correlation at T_c than any other cubic equation.

The Redlich-Kwong equation of state, in the form popularized by Soave (1972), is

$$P = \frac{RT}{v - b_{rk}} - \frac{a_{rk}(T)}{v(v + b_{rk})} \quad (29)$$

where

$$a_{rk} = 0.42748 \cdots \left(\frac{R^2 T_c^2}{P_c} \right) \cdot \alpha \quad (30)$$

$$b_{rk} = 0.08664 \cdots \left(\frac{RT_c}{P_c} \right) \quad (31)$$

α introduces the temperature dependence in $a_{rk}(T)$; in the original equation (Redlich and Kwong, 1949), it is given by

$$\alpha = 1/T_r^{0.5} \quad (32)$$

When eqn. (29) is combined with the B correlation (eqn. (10)) we get the virial-RK equation of state.

$$P = \frac{RT}{v - b_{rk}} - \frac{(b_{rk} - B)RT}{v(v + b_{rk})} \quad (33)$$

where

$$B = b_{rk} - a_{rk}(T)/RT \quad (34)$$

Prediction of C

Equation (33) now reproduces the B as given by eqn. (10), but can also predict C (and higher virial coefficients):

$$C = b_{rk}^2 + b_{rk} a_{rk}(T)/RT \quad (35a)$$

or

$$C = b_{rk}(2b_{rk} - B) \quad (35b)$$

Figure 6 gives two examples of C predictions with the virial-RK equation. The predictions for methane are in fair agreement with the experimental data. However, in the case of water, virial-RK predicts positive values, whereas the experimental data are strongly negative (the change in scale should also be noted). This failure of virial-RK is a serious one, but, most

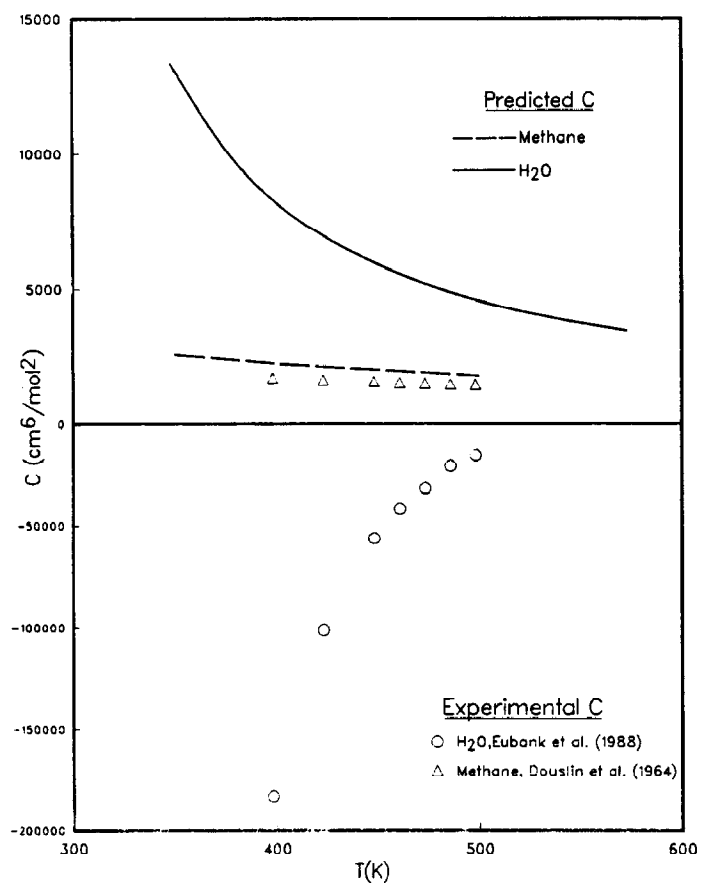


Fig. 6. Virial-RK prediction of third virial coefficients.

interestingly, is very similar to that of the Stockmayer potential (Hirschfelder et al., 1964). In spite of the poor results for the C of water, the next section shows that virial-RK can be used successfully to predict the fugacity coefficient of water in methane at high pressures.

Mixing rules

The mixing rules in the cubic equations of state provide formulas for predicting the a_{rk} and b_{rk} of mixtures. The most common or "classical" mixing rules are the one-fluid van der Waals mixing rules:

$$a_{rk,m} = \sum_i \sum_j y_i y_j a_{rk,ij} \quad (36)$$

$$b_{rk,m} = \sum_i \sum_j y_i y_j b_{rk,ij} \quad b_{rk,ij} = 0.5(b_{rk,i} + b_{rk,j}) \quad (37)$$

In the case of the virial-RK equation, we have two choices for $a_{rk,ij}$. If we treat it as a virial equation, then

$$a_{rk,ij} = (b_{rk,ij} - B_{ij})RT \quad (38)$$

and the value of $a_{rk,ij}$ depends on B_{ij} , which in turn depends on the value of the characteristic binary constant k_{ij} . In this approach, therefore, we can use the k_{ij} values determined by regressing B_{ij} data; for example, the values in Table 1 or Fig. 4.

Joffe (1978) made extensive calculations with eqn. (38), especially on water/alkane systems. He confirmed that the k_{ij} s from B_{ij} analysis gave good results, although in some cases it was noted that slightly lower k_{ij} s gave better results at high pressures. For example, for water/ethane, $k_{ij} = 0.37$ was best at low pressures, but at $P > 200$ bar the optimum value was $k_{ij} = 0.33$. With that value, the fugacity coefficient of water in ethane predicted with eqn. (33) was only 4% too high at 414 bar, while the original Redlich-Kwong equation was 67% below the experimental value (Reamer et al., 1943). Joffe also found eqn. (33) to be a significant improvement over the equations of state of de Santis et al. (1974) and Nakamura et al. (1976).

The second choice for $a_{rk,ij}$ is to treat virial-RK as a cubic equation of state. Then we introduce the binary constant C_{ij} to correct for the deviation of $a_{rk,ij}$ from the geometric mean.

$$a_{rk,ij} = (a_{rk,i} a_{rk,j})^{0.5} (1 - C_{ij}) \quad (39)$$

where

$$a_{rk,i} = (b_{rk,i} - B_i)RT \quad (40)$$

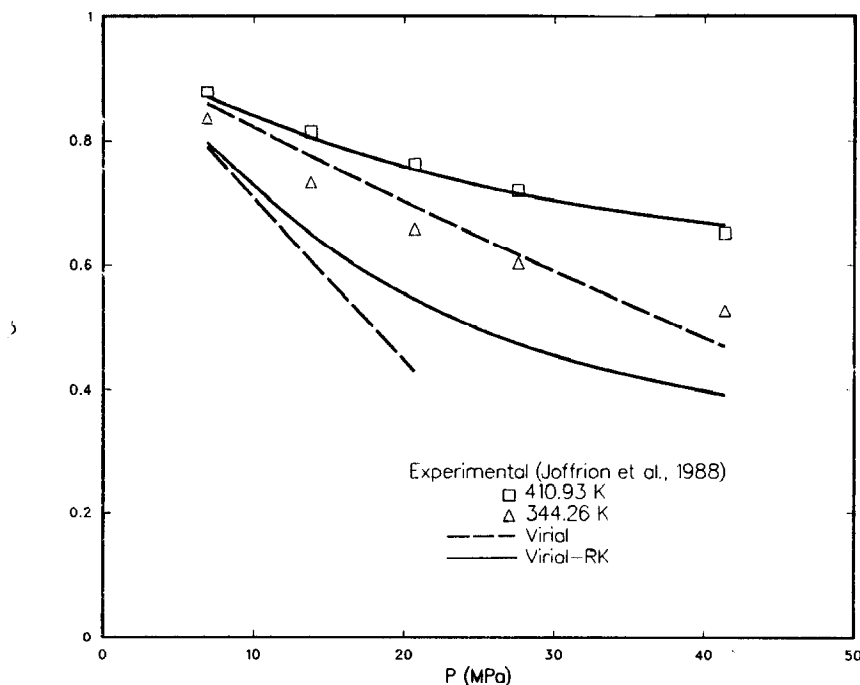


Fig. 7. Fugacity coefficient of water in methane.

The two approaches for $a_{rk,ij}$ produce similar results, although the values of k_{ij} and C_{ij} differ. For example, the optimum binary constants for water/methane are $k_{ij} = 0.34$ and $C_{ij} = 0.57$.

Figure 7 presents calculations and experimental data for the fugacity coefficient of water in methane. At 410.93 K, the virial-RK matches closely the data of Joffrion and Eubank (1988) up to 40 MPa, where the truncated virial equation is about 30% too low. However, at 344.26 K the virial-RK is much less satisfactory (25% too low at 40 MPa). As shown, the truncated virial equation is considerably worse and breaks down above 20 MPa. Not shown are the results with the original Redlich-Kwong equation, which is in error by more than 30%.

In spite of the discrepancy at 344.26 K (which may be due to experimental uncertainties related to the very low water levels), Fig. 7 supports the conclusion that virial-RK can be used to predict vapor-phase fugacity coefficients up to much higher pressures than with the truncated virial equation—and for systems that are more polar than with the original Redlich-Kwong equation of state.

CONCLUDING REMARKS

The B correlation provides a simple framework for correlating and even predicting the B of pure compounds and their mixtures. The polar contribution $f^{(2)}$ goes to zero as T_r becomes very large (actually, it approaches zero very rapidly for $T_r > 1$) or as μ_r goes to zero. For non-dimerizing polar organic compounds, eqns. (15) and (16) provide reliable estimates of a . Furthermore, these two equations merge into a single a vs. μ_r relation at high μ_r values (see Fig. 1).

When new, reliable B data become available, they can readily be used to determine the value of a (or of a and b for dimerizing compounds). Such new results were presented for water (Figs. 2 and 3). Also, new results were presented for water/*n*-alkane binaries, for which the characteristic binary parameter k_{ij} was successfully correlated with the critical volume of the hydrocarbon (Fig. 4). Such correlations (see also Tsonopoulos, 1979; Tsonopoulos et al., 1989) are useful in screening new B_{ij} data, as well as in predicting k_{ij} values when no B_{ij} data are available. Furthermore, these correlations can also be used with the virial-RK equation of state.

Equation (33), the virial-RK equation of state, is recommended for the calculation of vapor-phase fugacity coefficients, as well as of other vapor-phase thermodynamic properties. Its use can extend the applicability of hybrid models (where the liquid-phase non-ideality is accounted for through activity coefficients) up to very high pressures, even for highly polar systems. Equation (33) is limited to the vapor phase and should not be used to calculate liquid-phase fugacity coefficients and K values (with eqn. (3); see however, Kubic, 1982).

The failure of the virial-RK equation in the prediction of the third virial coefficient of water (see Fig. 6) will be examined in a separate publication. As already noted, however, very little is known about third virial coefficients of polar systems. Even for non-polar gases, the predicted C is reasonable only for $T_r > 0.9$, because eqn. (35) does not predict a maximum at $T_r \approx 0.9$.

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LIST OF SYMBOLS

a, b	parameters of polar contribution term to B , $f^{(2)}$ (eqn. (13))
a_{rk}, b_{rk}	Redlich-Kwong equation of state parameters

B	second virial coefficient
C	third virial coefficient
C_{ij}	binary interaction parameter in eqn. (39)
f_i	fugacity of component i
$f^{(0)}, f^{(1)}, f^{(2)}$	dimensionless terms of eqn. (10)
k_{ij}	binary interaction parameter in eqn. (19)
K_i	y_i/x_i ; equilibrium ratio of component i
P	pressure
R	gas constant
T	temperature
v	molar volume
x_i	liquid mole fraction of component i
y_i	vapor mole fraction of component i
Z	compressibility factor $= Pv/RT$

Greek letters

α	temperature dependence of Redlich–Kwong equation of state parameter a_{rk}
γ_i	activity coefficient of component i
μ	dipole moment
ϕ_i	fugacity coefficient of component i
ω	acentric factor

Subscripts

c	critical property
i, j	property of component i, j
ij	property of i – j interaction
m	mixture property
r	reduced property
rk	Redlich–Kwong equation of state parameters

Superscripts

L	liquid-phase property
V	vapor-phase property
0	reference property
'	virial coefficients in pressure series (eqn. (28))
∞	infinite-dilution property

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